ECE 329
Lecture 15

Magnetic Vector Potential
Inductance

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Magnetic Vector Potential $\vec{A}$

$$\nabla \bullet \vec{B} = 0$$ Gauss' Law

If the divergence is zero, then **B** can be written as the curl of a vector

New Definition: Magnetic Potential Vector

$$\vec{B} = \nabla \times \vec{A}$$

Oddly familiar:

$$\nabla \bullet \left(\nabla \times \vec{A} \right) = 0$$

Faraday's Law

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\frac{\partial}{\partial t}(\nabla \times \vec{A})$$

$$\nabla \times \vec{E} = -\nabla \times \frac{\partial \vec{A}}{\partial t}$$

$$\nabla \times (\vec{E} + \frac{\partial \vec{A}}{\partial t}) = 0$$

If curl is zero, then can be written as the gradient of a scalar

$$\vec{E} + \frac{\partial \vec{A}}{\partial t} = -\nabla \Phi$$

Oddly familiar:

$$\nabla \times \left(\nabla \Phi \right) = 0$$

$E = -\nabla \phi$ static case.

Electric and Magnetic Potentials

$$\vec{E} = -\nabla\Phi - \frac{\partial \vec{A}}{\partial t}$$

Electric Potential Magnetic Potential

$$\vec{B} = \nabla \times \vec{A}$$

With these definitions, we automatically satisfy Faraday's Law & Gauss' Magnetic Law

Instead of 6 unknowns: (E_x, E_y, E_z) & (B_x, B_y, B_z)
we have 4: (A_x, A_y, A_z) and V/ϕ

Gauss' Electric Law for Potentials

$$\nabla \bullet \vec{D} = \rho$$

$$\nabla \bullet \varepsilon \vec{E} = \rho$$

$$\nabla \bullet \vec{E} = \frac{\rho}{\varepsilon}$$

Assuming ε is constant

$$\nabla \bullet \left(-\nabla \Phi - \frac{\partial \vec{A}}{\partial t} \right) = \frac{\rho}{\varepsilon}$$

1 equation

Ampere's Law for Potentials

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \times \frac{\vec{B}}{\mu} = \vec{J} + \varepsilon \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{B} = \mu \vec{J} + \mu\varepsilon \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{B} - \mu\varepsilon \frac{\partial \vec{E}}{\partial t} = \mu \vec{J}$$

$$\nabla \times \left(\nabla \times \vec{A}\right) - \mu\varepsilon \frac{\partial}{\partial t}\left(-\nabla\Phi - \frac{\partial \vec{A}}{\partial t}\right) = \mu \vec{J}$$

3 equations

Special Case: Static Fields

Gauss

$$\nabla \bullet \left(-\nabla\Phi - \frac{\partial \vec{A}}{\partial t} \right) = \frac{\rho}{\varepsilon}$$

$\frac{d}{dt} \to 0$.

$$\nabla \bullet \left(\nabla\Phi\right) = -\frac{\rho}{\varepsilon}$$

We studied this one already (Poisson):
Relationship between a charge distribution and the potential field

Ampere

$$\nabla \times \left(\nabla \times \vec{A}\right) - \mu\varepsilon \frac{\partial}{\partial t}\left(-\nabla\Phi - \frac{\partial \vec{A}}{\partial t} \right) = \mu\vec{J}$$

$$\nabla \times \left(\nabla \times \vec{A}\right) = \mu\vec{J}$$

Challenge Question: Gauge Transformation

- Suppose **E** and **B** can be represented by the scalar and vector potentials: Φ and **A**.

$$\vec{E} = -\nabla\Phi - \frac{\partial \vec{A}}{\partial t} \qquad \vec{B} = \nabla \times \vec{A}$$

Which of the following is true:

(a) Φ and **A** are not uniquely defined

(b) $\vec{A}' = \vec{A} + \nabla\lambda,\ \Phi' = \Phi - \frac{\partial \lambda}{\partial t}$ also represents **E** and **B**

(c) The divergence of **A** must be 0

(d) The laplacian of Φ must be $-\rho/\varepsilon$

Not unique

$\phi' = \phi + 2V \rightarrow E' = -\nabla\phi' - \frac{\partial A}{\partial t}$

$= -\nabla\phi - \frac{\partial A}{\partial t}$

$\lambda(x,y,z,t)$

free to choose $\nabla \cdot \vec{A}$

only static case

(b) $\vec{E} = -\nabla\left(\phi - \frac{\partial \lambda}{\partial t}\right) - \frac{\partial}{\partial t}(A + \nabla\lambda)$

$= -\nabla\phi + \nabla\frac{\partial \lambda}{\partial t} - \frac{\partial A}{\partial t} - \frac{\partial}{\partial t}\nabla\lambda$

cancel out

$\vec{B} = \nabla \times (\vec{A} + \nabla\lambda) = \nabla \times \vec{A} + \nabla \times (\nabla\lambda)$, $\nabla \times (\nabla\lambda) = 0$

Lecture 15a Summary

- Since div curl **A** = 0 and curl grad f =0,

$$\vec{E} = -\nabla\Phi - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A}$$

satisfy Faraday's and Gauss' Mag. Laws

- For static fields, Poisson's equation is

$$\nabla^2\Phi = -\frac{\rho}{\varepsilon}$$

ECE 329
Lecture 15b

Inductance

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Inductance

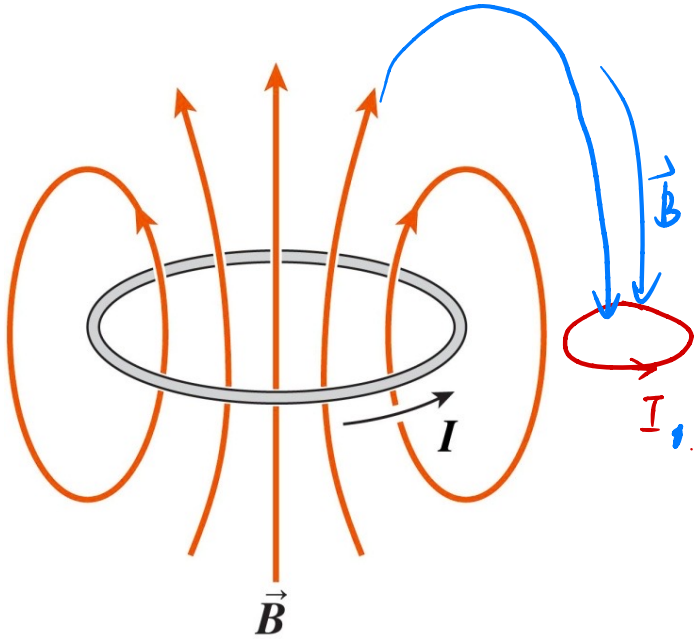
Self-inductance occurs when a changing current in a circuit results in an induced emf that opposes the change in the circuit itself.

- Self-inductance occurs because some of the magnetic flux produced in a circuit passes through that same circuit.

Flux-current relation $\Psi = LI$

➡ Self-inductance $L = \dfrac{\Psi}{I}$ [H]

The unit of inductance is the **henry**



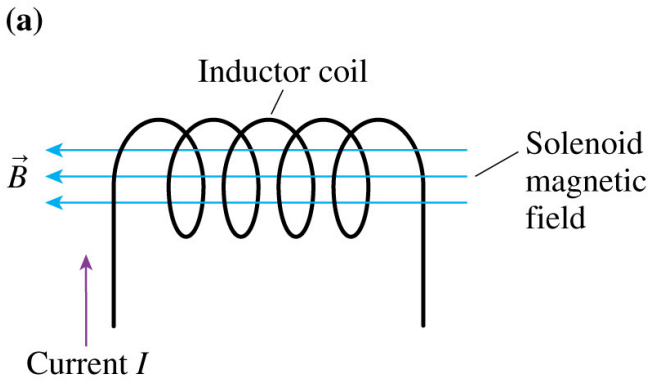
Induced emf

The inductance of the wire creates an emf that opposes rapid changes in the current

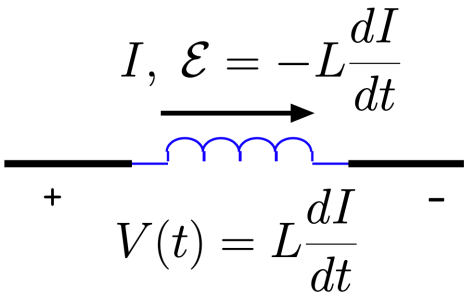
Faraday's Law

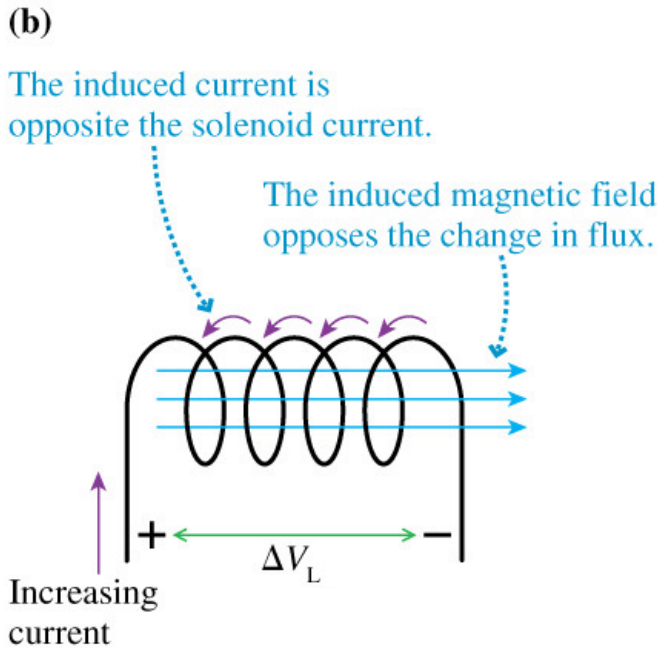
$$emf = \oint \vec{E} \bullet d\vec{l} = -\frac{d\psi_B}{dt}$$

$$emf = -\frac{d(LI)}{dt} = -L\frac{dI}{dt}$$



Circuit symbol



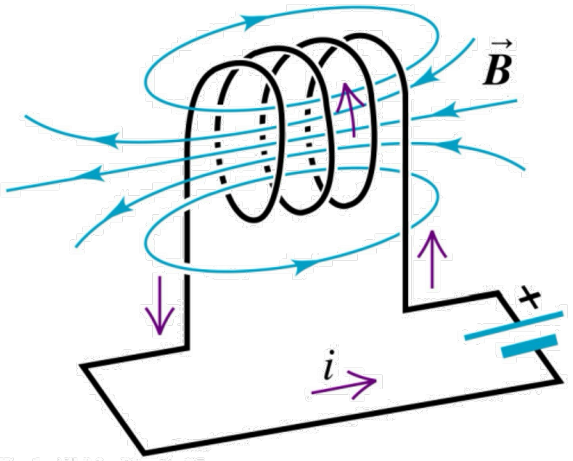


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Self-Inductance and Inductors

For an inductor consisting of N-loops, the emf $\mathcal{E}$ measured around N-loops is

$$\mathcal{E} = -N\frac{d\Psi}{dt} = L\frac{dI}{dt}$$



➡ Self-inductance

$$L \equiv N L_{single}$$

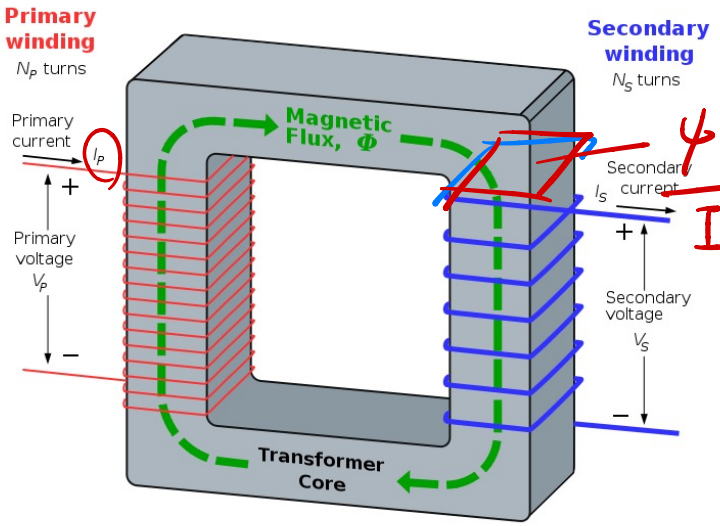
A device designed to have a particular inductance L is called an inductor



Examples from http://upload.wikimedia.org/wikipedia/commons/2/27/Aplikimi_i_feriteve.png

Many applications happen in AC circuits

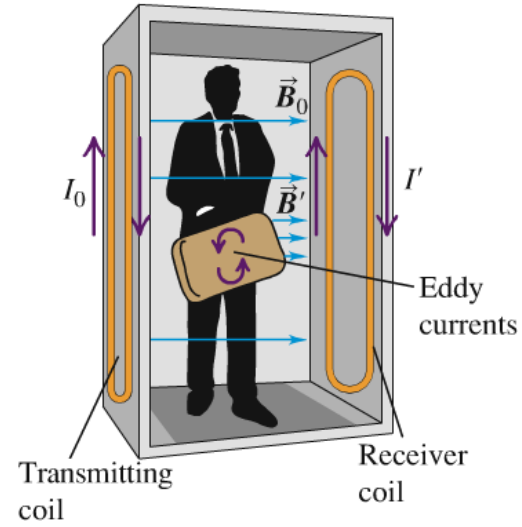
Transformers

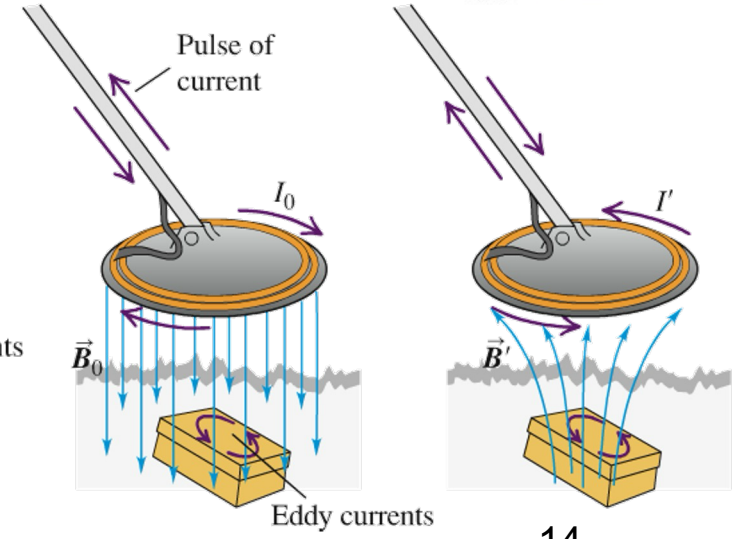




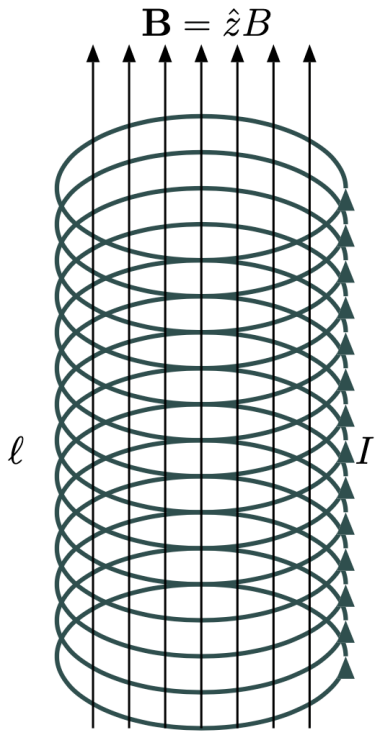
Metal detectors







Example: Find the inductance of an infinitely long solenoid



density: N loops per unit length

$\vec{B}$ in the interior of the solenoid is

$$\vec{B} = \mu_0 I N \hat{z}$$

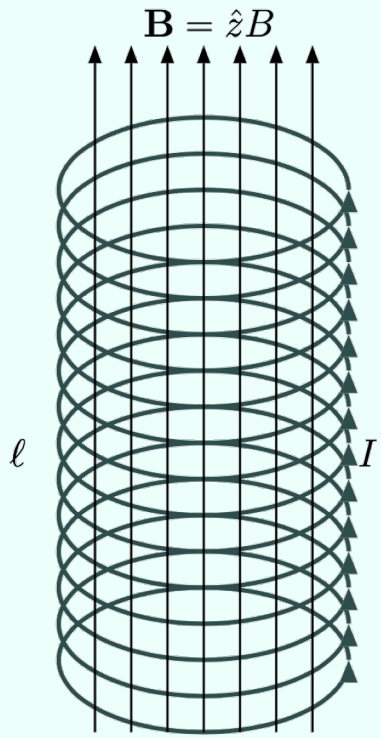
the number of turns of the solenoid is

$$n = N\ell$$

the inductance of the solenoid with n turns is

$$L = \frac{n\Psi}{I} = \frac{N\ell(\mu_0 I N)A}{I} = N^2 \mu_0 A \ell$$

Stored Energy in Infinite Solenoid



Infinite Solenoid

$$\vec{\boldsymbol{B}} = \mu_0 I N \hat{z}$$

$$L = N^2 \mu_0 A \ell$$

Power in inductor

$$P = VI = L\frac{dI}{dt}I = \frac{d}{dt}\left(\frac{1}{2}LI^2\right)$$

Stored energy is

$$W = \frac{1}{2}LI^2 = \frac{1}{2}N^2\mu_0 A\ell I^2$$

$$= \frac{|B_z|^2}{2\mu_0}A\ell = \frac{1}{2}\mu_0|H_z|^2 A\ell$$

$L \to N^2 \mu_0 A\ell$

$\vec{B} \to \mu_0 N I \hat{z}$

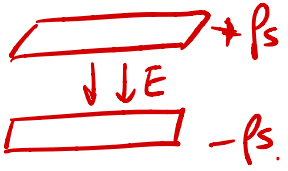
$A\ell$ → Volume of solenoid

Stored magnetostatic energy per unit volume is

$$w = \frac{1}{2}\mu_0|H_z|^2 = \frac{1}{2}\mu_0 \vec{\boldsymbol{H}} \cdot \vec{\boldsymbol{H}}$$

Q: What's the stored electrostatic energy per unit volume in a parallel plate capacitor?

$$w = \frac{1}{2}\varepsilon_0 \vec{E} \cdot \vec{E}$$



Steps for Finding Inductance

- Find H(r)
- Find B(r)
- Find Magnetic flux by integrating
- Inductance
- Per unit length inductance

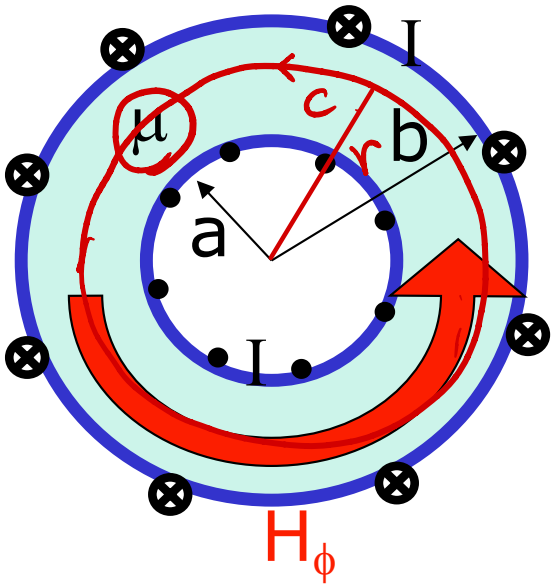
④ $L = \frac{\psi_B}{I} = \frac{③ \iint \vec{B} \cdot d\vec{s}}{I}$

② $\vec{B} = \mu \vec{H}$

① $\oint \vec{H} \cdot d\vec{l} = I_{enc}$

⑤ $\mathcal{L} = \frac{L \rightarrow \text{inductance}}{l \leftarrow \text{length}}$

Inductance of a Coax Cable



Current I flows down the length of the outer conductor and returns in the opposite direction through the inner conductor

Step 1: Find H(r)

$$\oint_C \vec{H} \cdot d\vec{\ell} = I_{enc}$$

Ampere's Law

$$H \cdot 2\pi r = I$$

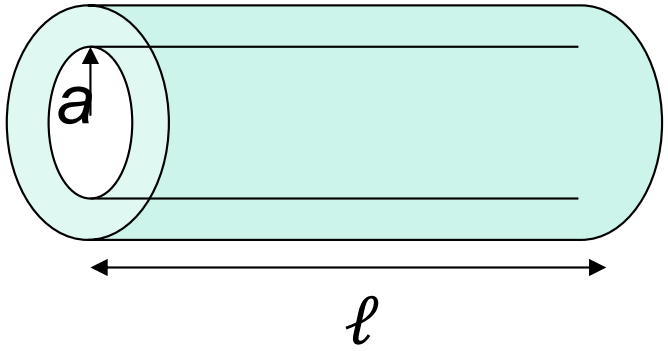
$$H = \frac{I}{2\pi r}\hat{\phi}$$

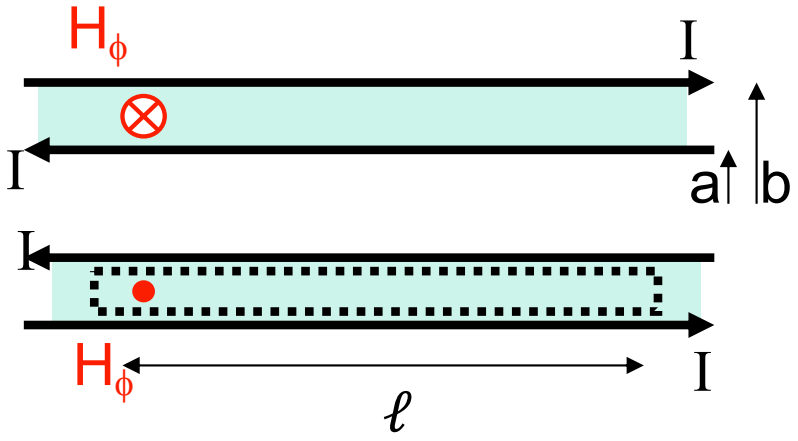
So the magnetic field is

$$H_\phi = \frac{I}{2\pi r} \quad A/m$$

in between the coax

Inductance of Coaxial Cable





Step 2: Find B(r)

$$\vec{B} = \mu\vec{H} = \frac{\mu I}{2\pi r}\hat{\phi}$$

[T or Wb/m]

Step 3: Find Ψ

$$\Psi = \int_S \vec{B}\cdot d\vec{S} = \int_{z=0}^{\ell}\int_{r=a}^{r=b}\frac{\mu I}{2\pi r}\hat{\phi}\cdot\hat{\phi}\,dr\,dz$$

$$= \frac{\mu I \ell}{2\pi}\int_a^b \frac{1}{r}dr = \frac{\mu_0 I \ell}{2\pi}\ln(b/a) \quad [Wb]$$

Ψ: Magnetic Flux $[Wb]$

$\vec{B}$: Magnetic Flux Density $\left[\frac{Wb}{m^2}\right]$

Inductance

Step 4: Find inductance L

$$\mathrm{L} = \frac{\Psi}{\mathrm{I}}$$

$$\mathrm{L} = \frac{\mu \ell}{2\pi} \ln(b/a)$$ Units: Henry (H)

Step 5: Find per unit length inductance $\mathcal{L}$

$$\mathcal{L} = \frac{L}{\ell} = \frac{\mu}{2\pi} \ln(b/a)$$ Inductance/Length (H/m)

Relationships between Capacitance, Conductance & Inductance

	Capacitance per unit length $\mathcal{C}$	Conductance per unit length $\mathcal{G}$	Inductance per unit length $\mathcal{L}$
Parallel plate conductors	$\mathcal{C} = \epsilon \frac{w}{d}$ G.F.	$\mathcal{G} = \sigma \frac{w}{d}$	$\mathcal{L} = \mu \frac{d}{w}$
Coaxial cables	$\mathcal{C} = \frac{2\pi\epsilon}{\ln(b/a)}$ G.F.	$\mathcal{G} = \frac{2\pi\sigma}{\ln(b/a)}$	$\mathcal{L} = \frac{\mu}{2\pi}\ln(b/a)$

$$\mathcal{C} = \epsilon \cdot Geometrical_Factor$$

$$\mathcal{G} = \sigma \cdot Geometrical_Factor$$

$$\mathcal{L} = \mu / Geometrical_Factor$$

This is true in general for any pair of infinitely long, parallel perfect conductors and so we have the following:

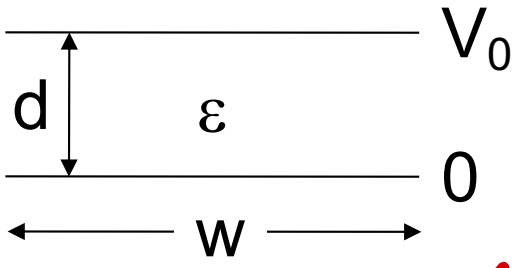
$v_p = \frac{1}{\sqrt{\mu\epsilon}}$

$$\mathcal{L}\mathcal{C} = \mu\epsilon$$ and $$\frac{\mathcal{G}}{\mathcal{C}} = \frac{\sigma}{\epsilon}$$

$v_p = \frac{1}{\sqrt{LC}}$ $C = \frac{1}{\sqrt{\mu\epsilon_0}}$.

Challenge Question
Parallel plate capacitor

- For the parallel plate capacitor, we found



$$C = \epsilon \frac{A}{d} \Rightarrow \mathcal{C} = \epsilon \frac{w}{d}$$

If the plate separation d increases, which is true:

(a) $\mathcal{G}$ and $\mathcal{L}$ will both increase

(b) $\mathcal{C}$ and $\mathcal{G}$ will both increase

(c) $\mathcal{L}$ will increase, but $\mathcal{C}$ will decrease

(d) $\mathcal{G}$ will increase, but $\mathcal{L}$ will decrease

$\mathcal{L}\mathcal{C} = \text{const} = \mu\varepsilon$

$\frac{\mathcal{C}}{\mathcal{G}} = \frac{\varepsilon}{\sigma} = \text{const}$

Lecture 15 Summary

- Capacitance $C = Q/V_0$
- Conductance $G = |I_c|/V_0$
- Inductance $L = \psi/I$
- Relationships $\mathcal{LC} = \mu\epsilon$ and $\frac{\mathcal{G}}{\mathcal{C}} = \frac{\sigma}{\epsilon}$
- Next Up
  - Conservation of Charge